Longevity orthologs across four major model organisms

Complete visual catalogue • 320 genes • 206 direct pairs

Reviewed example = a checked experiment; GenAge annotation = catalogue evidence only.

All genes in the qualifying pair set are included; rows organise direct links, not proven conserved mechanisms.

	Budding yeast	Nematode	Fruit fly	Mouse
C035 **GPA2** 6 genes / 2 species 5 direct pairs	**Y1 GPA2** GenAge 319 **GenAge annotation** **Anti-Longevity** Lifespan: Increase + Decrease Refs: 11000115, 20421943	**W1 egl-30** GenAge 919 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 19043554, 20647387 **W2 goa-1** GenAge 942 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 19043554 **W3 gpa-3** GenAge 314 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 17187771 **W4 gsa-1** GenAge 943 **GenAge annotation** **Pro-Longevity** Lifespan: Decrease Refs: 18059442 **W5 odr-3** GenAge 481 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17187771, 20332814	No gene in this candidate network	No gene in this candidate network

Direct pairs: W1–Y1 (n:m); W2–Y1 (n:m); W3–Y1 (n:m); W4–Y1 (n:m); W5–Y1 (n:m)

C037 **GPX2** 2 genes / 2 species 1 direct pair	**Y1 GPX2** GenAge 1578 **GenAge annotation** **Necessary for fitness** Lifespan: Decrease Refs: 19030232	No gene in this candidate network	No gene in this candidate network	**M1 Gpx4** GenAge 324 **GenAge annotation** **Anti-Longevity** Lifespan: Increase Refs: 17895430

Direct pairs: M1–Y1 (n:m)

Y/W/F/M identify genes in yeast/worm/fly/mouse columns. Pair codes retain the OMA source orientation.
Refs are catalogue-linked PMIDs, not newly reviewed experiments. Empty cells do not establish biological absence.
GenAge lifespan labels do not specify interventions. Only reviewed examples have experiment arrows and colours.
Sources: GenAge Build 21; OMA All.May2026. See atlas legend, gene index and source tables for evidence details.